

Verifai GRC - Razorpay Payment Integration

Complete Guide to Autopay Subscriptions

1. What is This Document About?

This document explains how payments work in Verifai GRC. We use **Razorpay Autopay** to automatically collect subscription payments from customers.

Who should read this?

- Business teams wanting to understand payment flows
- Support teams handling customer payment queries
- Developers maintaining or extending the payment system

2. How Our Subscription Works - The Big Picture

2.1 The 2-Year Subscription Model

When a customer signs up for Verifai GRC, they commit to a **2-year subscription** with the following structure:

Period	Duration	Price per Module	What Happens
Trial	14 days	FREE	Customer uses the platform for free
Year 1	12 months	₹1,200/year (₹100/month)	Promotional pricing
Year 2	12 months	₹1,80,000/year (₹15,000/month)	Standard pricing

Example: If a customer signs up for the GRC module:

- Day 1-14: Free trial
- Day 15: ₹1,416 charged (₹1,200 + 18% GST) for Year 1
- Day 380: ₹2,12,400 charged (₹1,80,000 + 18% GST) for Year 2

2.2 What is Autopay?

Autopay (also called "mandate") is an authorization that allows us to automatically deduct money from the customer's bank account or card on scheduled dates.

Supported Payment Methods:

Method	How it Works	Common Apps
UPI AutoPay	Customer approves via UPI app	GPay, PhonePe, Paytm, BHIM
e-NACH	Bank mandate authorization	All major banks
Card Mandate	Card saved for recurring charges	Visa, Mastercard

3. The Complete Customer Journey

Step-by-Step: What Happens When Someone Signs Up

Day 1: Sign Up & Authorize Payment

CUSTOMER SIGNS UP

1. Customer fills signup form:

• Organization name

• Admin name & email

• Password

• Selects modules (GRC, TPRM, Internal Audit)

2. Customer agrees to:

✓ 2-year subscription commitment

✓ Autopay authorization

3. Razorpay checkout opens:

• Customer selects payment method (UPI/Card/Bank)

• Authorizes the mandate (recurring payment permission)

• Small verification charge of ₹2 (refunded)

4. Mandate authorized → Trial begins immediately!

Day 1-14: Free Trial Period

During the 14-day trial:

- Customer has FULL access to all selected modules
- No charges are made
- Customer can explore and set up their organization
- The mandate (payment authorization) is active but unused

Day 15: First Automatic Payment

TRIAL ENDS - FIRST CHARGE

What happens automatically on Day 15:

1. Razorpay initiates the charge using the stored mandate
2. Customer's bank/UPI app processes the payment
(Customer does NOT need to do anything)
3. If successful:
 - ✓ Invoice marked as PAID
 - ✓ Subscription status → ACTIVE
 - ✓ Customer continues using the platform
4. If failed:
 - x Razorpay retries up to 3 times over 7 days
 - x If all retries fail → Subscription SUSPENDED
 - x Customer loses access until payment is resolved

Year 2: Automatic Renewal

On the anniversary of the subscription (Day 380), the Year 2 charge happens automatically:

Event	What Happens
7 days before	System sends reminder email
On due date	Razorpay charges the stored mandate
If successful	Subscription continues for another year
If failed	Retries attempted, then suspension

4. Payment States Explained

4.1 Mandate (Authorization) States

State	What It Means	Customer Access
Pending	Customer hasn't completed authorization yet	No access
Authenticated	Mandate authorized, trial active	Full access (trial)
Active	First payment captured, subscription running	Full access
Halted	Payment failed after all retries	ACCESS REVOKED
Cancelled	Customer or admin cancelled	ACCESS REVOKED
Completed	All contract charges done (2 years)	Access until expiry

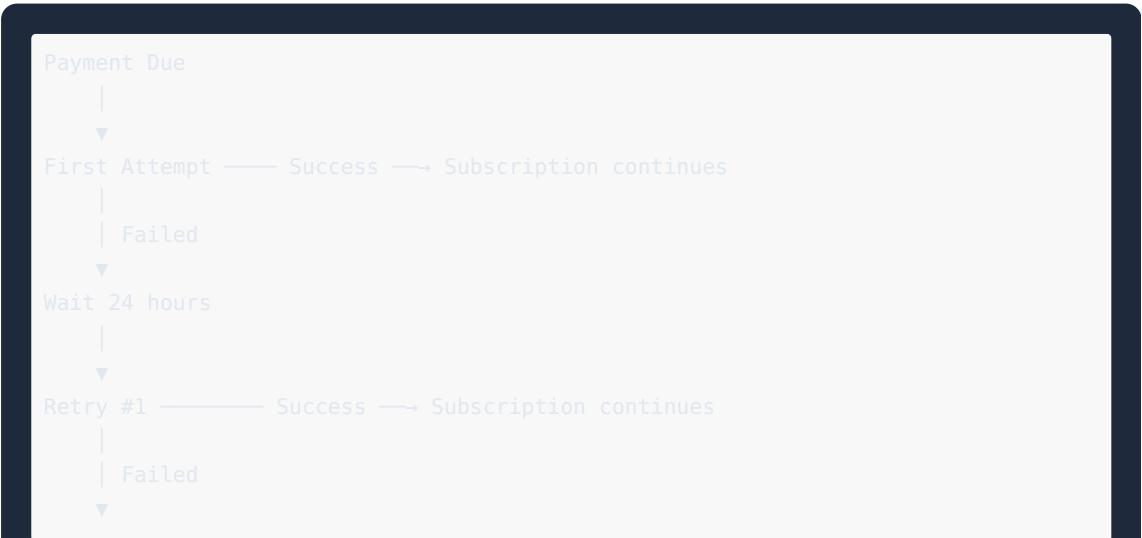
4.2 What Happens When Payment Fails?

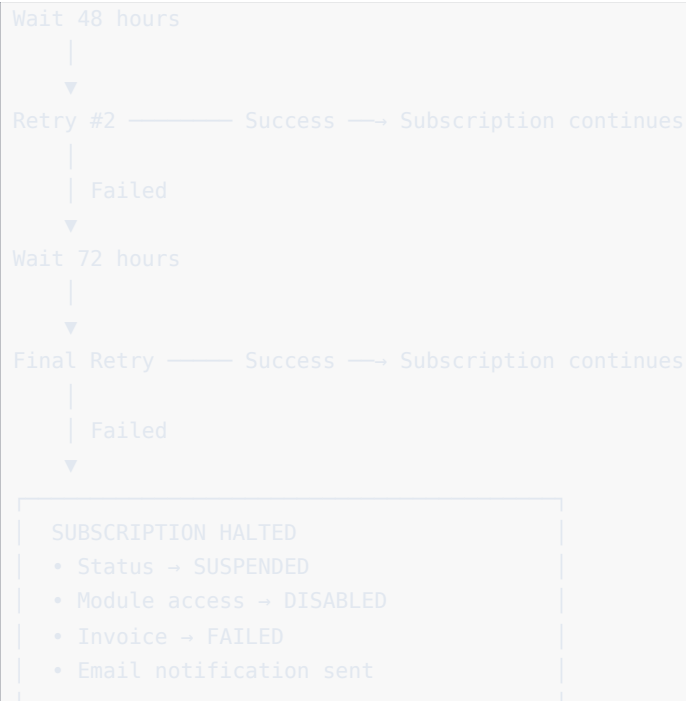
IMPORTANT: Our Policy on Failed Payments

Because this is a mandatory 2-year contract with autopay, if autopay fails or is disabled, the subscription ends **immediately**.

This is by design - the contract requires working autopay authorization.

The Failure Sequence:





5. How Razorpay Communicates With Us

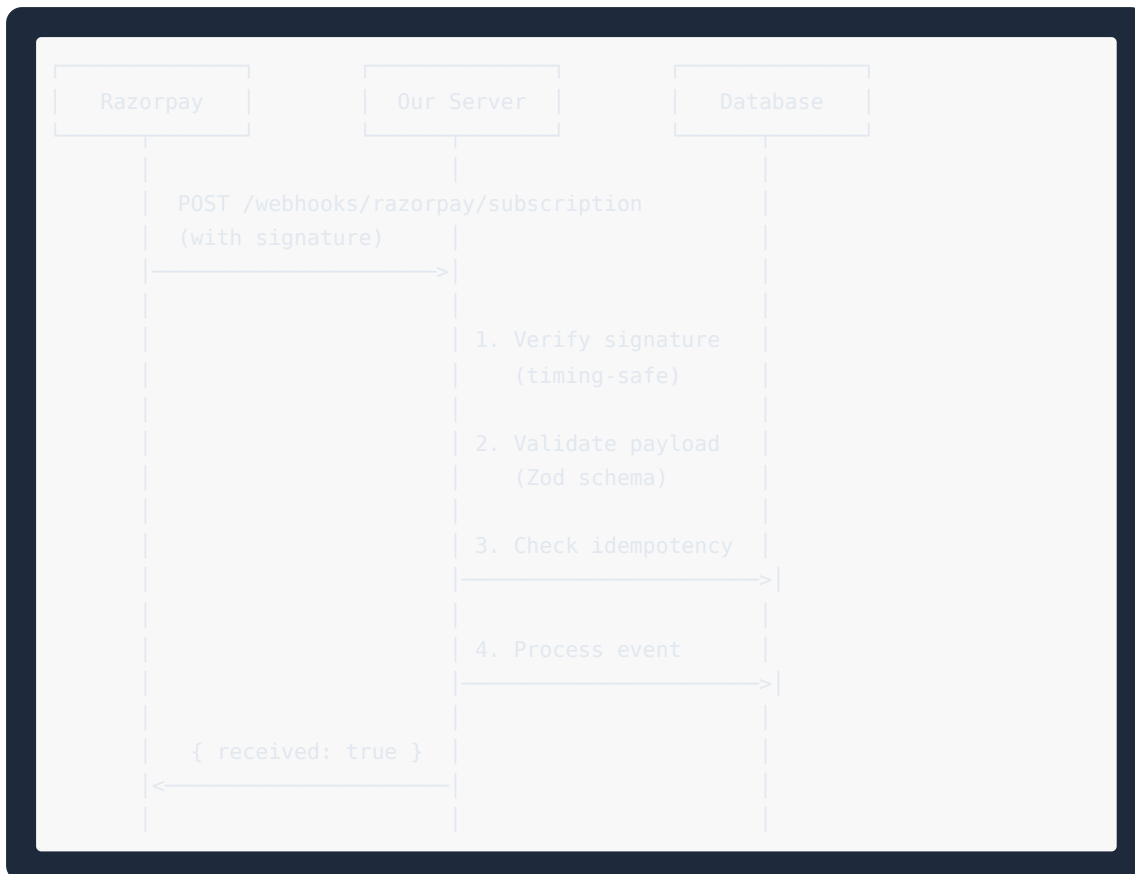
5.1 Webhooks - Real-Time Notifications

Razorpay sends us instant notifications (called "webhooks") whenever something happens with a payment or subscription.

Key Events We Handle:

Event	When It Fires	What We Do
subscription.authenticated	Customer authorizes mandate	Start trial, enable access
subscription.charged	Payment successfully captured	Mark invoice PAID, keep access
subscription.halted	All payment retries failed	Suspend subscription, revoke access
subscription.cancelled	Subscription cancelled	End subscription, revoke access
subscription.completed	All 2-year charges done	Mark as completed
payment.failed	Individual payment attempt failed	Log error, wait for retry

5.2 The Webhook Flow (Technical)



6. Security Measures

6.1 How We Protect Payments

Our security follows Razorpay's official best practices and adds extra protection.

Signature Verification

Every message from Razorpay includes a digital signature. We verify it using a **timing-safe comparison** to prevent hackers from guessing the signature.

```
// We use crypto.timingSafeEqual() - not simple === comparison
// This prevents "timing attacks" where hackers measure response times
```

Payload Validation

We validate every webhook payload against a strict schema:

- Event type must be present
- Required fields must exist
- Data types must match expected formats

If validation fails → We reject the webhook with error 422.

Fail-Closed Security

In production, if security credentials are missing:

- We **REJECT** the webhook (not accept)
- This prevents processing of potentially forged requests

Idempotency Protection

We track every webhook event ID in the database:

- If we receive the same event twice → Process only once
- Prevents duplicate charges or duplicate processing

7. Technical Implementation Details

7.1 Key Files in the Codebase

File	Purpose
<code>src/lib/payment-provider.ts</code>	Core Razorpay client, signature verification
<code>src/lib/payment-provider-mandate.ts</code>	Create/manage subscription mandates
<code>src/lib/razorpay-plan-manager.ts</code>	Manage Razorpay plans
<code>src/app/api/webhooks/razorpay/subscription/route.ts</code>	Handle subscription webhooks
<code>src/app/api/webhooks/razorpay/route.ts</code>	Handle payment webhooks

7.2 Database Tables

Table	Purpose
<code>RazorpayPlan</code>	Cached Razorpay plan IDs
<code>RazorpayEvent</code>	Webhook idempotency tracking
<code>ModuleSubscription</code>	Links modules to mandate IDs
<code>Subscription</code>	Customer subscription records

Invoice	Billing records
Payment	Payment transaction records

7.3 Environment Variables

```
# Required for Razorpay integration
RAZORPAY_KEY_ID=rzp_live_XXXXX # API Key ID
RAZORPAY_KEY_SECRET=XXXXX # API Secret (NEVER expose publicly)
RAZORPAY_WEBHOOK_SECRET=XXXXX # Webhook signature secret

# Configuration
PAYMENT_STUB=false # Set true for testing without real payments
TRIAL_DAYS=14 # Trial period duration
```

8. Common Scenarios & Troubleshooting

8.1 Customer Can't Complete Payment Authorization

Possible causes:

1. Bank doesn't support UPI AutoPay
2. Customer cancelled during checkout
3. Network timeout

Solution: Customer should retry with a different payment method or contact their bank.

8.2 Automatic Payment Failed

Possible causes:

1. Insufficient funds in customer's account
2. Card expired
3. Bank declined the transaction
4. Customer disabled autopay in their banking app

What happens:

- Razorpay retries automatically (up to 3 times)
- Customer receives notification emails
- If all retries fail → Subscription suspended

Solution: Customer needs to ensure sufficient funds and valid payment method, then contact support to retry.

8.3 Customer Wants to Cancel

Important: Cancellation is NOT available during the 2-year contract period. This is a binding commitment.

After 2 years, customer can choose not to renew by contacting support before the renewal date.

9. Pricing Summary

Per Module Pricing (+ 18% GST)

Module	Year 1 (Promo)	Year 2 (Standard)	2-Year Total
GRC	₹1,200/year	₹1,80,000/year	₹1,81,200
TPRM	₹1,200/year	₹1,80,000/year	₹1,81,200
Internal Audit	₹1,200/year	₹1,80,000/year	₹1,81,200

With GST (18%):

Module	Year 1 Total	Year 2 Total	2-Year Grand Total
GRC	₹1,416	₹2,12,400	₹2,13,816
TPRM	₹1,416	₹2,12,400	₹2,13,816
Internal Audit	₹1,416	₹2,12,400	₹2,13,816

10. Contact & Support

For payment-related issues:

- **Customer Support:** Support team can check payment status in admin panel
- **Technical Issues:** Check `RazorpayEvent` table for webhook processing errors
- **Razorpay Dashboard:** Access at <https://dashboard.razorpay.com>

Appendix A: Complete Timeline Example

Customer: ABC Corp signs up for GRC module on January 1, 2026

Date	Day	Event	Amount
Jan 1, 2026	1	Signup, mandate authorized, trial starts	₹2 (verification, refunded)
Jan 1-14, 2026	1-14	Free trial period	₹0
Jan 15, 2026	15	Year 1 charge (auto)	₹1,416
Jan 15, 2027	380	Year 2 charge (auto)	₹2,12,400
Jan 15, 2028	745	Contract ends	-

Total paid over 2 years: ₹2,13,816 (including GST)

Appendix B: Webhook Event Reference

Event	Trigger	Our Action
subscription.authenticated	Mandate authorized	Enable trial access
subscription.pending	Awaiting authorization	Show pending status
subscription.charged	Payment captured	Mark invoice PAID
subscription.halted	All retries failed	Suspend immediately
subscription.paused	Subscription paused	Treat as cancelled
subscription.cancelled	Cancelled by user/admin	Revoke access
subscription.completed	All charges done	Mark completed
payment.captured	Payment successful	Update payment record
payment.failed	Payment attempt failed	Log, await retry

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